Topic : Interactive fossil learning via phone AR quizzes

How can phone-based AR with knowledge-testing tasks (instead of static labels) turn fossil exhibits into interactive learning journeys, and to what extent does it boost young visitors’ engagement and knowledge retention compared with reading labels alone?

Specialization : Interactive Media

IT22249920| Insandu W M P S | Project ID: 25-26J-417

Reimagining Fossil Exhibits

Knowledge Gap / Problem Definition

- Fossil exhibits in Sri Lanka’s museums remain static and text-heavy, failing to engage youth.
- No fossil-specific AR systems exist locally, and current museum AR apps lack collaboration and gamification.

Shortcomings of Existing Systems

- International AR projects often provide single-user, passive experiences.
- Markerless AR tracking is expensive and unreliable for local conditions.

Proposed Solution

- QR code–anchored AR scavenger hunt for fossil fragments.
- Integration of quizzes and skeleton assembly to reinforce knowledge.
- Multi-user synchronization to encourage collaborative learning.
- Treasure map–style navigation to reduce distractions and guide exploration.

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Research Gap

Work / Features	Phone-based AR (no headset)	QR-guided / treasure-map navigation	Quiz-gated tasks (active recall)	Direct A/B vs. labels-only
Selected Paper 1 – Paliokas et al., 2020 (Gamified heritage AR)	✓	✓	✓	✗
Selected Paper 2 – Savela et al., 2020 (AR quiz field experiment)	✓	✗	✓	✓
Selected Paper 3 – Ch'ng et al., 2023 (Social AR in heritage)	✓	✗	✗	✗
Selected Paper 4 – Zhou et al., 2022 (Museum AR/VR meta-analysis)	✓	✗	◐*	◐*
▶ Interactive fossil learning via phone AR quizzes	✓	✓	✓	✓

System Gap

Product / Features	**Phone-based AR** *(no headset)*	**QR-code anchoring**	**Guided navigation** *(treasure-map)*	**Quiz-gated unlocking / tasks**
Merge EDU	✓	✗ *(uses Merge Cube)*	✗	✓
CoSpaces Edu	✓	◐ *(markerless / MergeCube)*	✗	✓ *(teacher-made)*
GeoQuest AR	✓	✗ *(GPS/location)*	✓	✗
Wonderscope (story AR)	✓	✗	✗	✗
▶ Interactive fossil learning via phone AR quizzes	✓	✓	✓	✓

Functional Requirements

- Onboarding + walk-safe
- Treasure-map guidance (offline)
- QR scan → ExhibitID → anchor
- Fragment card (name, fact, lock)
- Quiz engine (1–3 Qs)
- Feedback + retry
- Inventory + progress + sync
- Skeleton assembly + completion
- Pre/Post mini-quizzes (A/B)
- Group Mode (code-based)
- Curator updates (Firebase/CSV)
- Accessibility (audio; SI/TA/EN)

Non Functional Requirements

- Performance
- Scalability
- Usability
- Immersion Quality
- Accessibility
- Reliability
- Compatibility
- Maintainability
- Cost-Efficiency

Application of Specialization-Based Knowledge

Domains of Knowledge Applied

- Human–Computer Interaction (HCI) – ensuring usability for young audiences.
- Augmented Reality Development – Unity 3D, AR Foundation.
- Game Design & Gamification – scavenger hunt, rewards, interactive challenges.
- Networking – Photon multiplayer sessions for real-time collaboration.

Technologies Used

- Unity 3D + AR Foundation (ARCore & ARKit support).
- Photon Networking for synchronized multi-user AR.
- Blender/3ds Max for 3D fossil fragment modeling & animation.
- QR Code libraries (Zxing/AR Foundation support)

Measuring Impact & Real-World Use

Evaluation & Success Criteria

- Quantitative: Quiz accuracy rates, fossil completion %, average exploration time.
- Qualitative: Visitor surveys, interviews, observed engagement levels.
- Target Outcome: At least 30% increase in engagement compared to static exhibits.

Data Availability & Ethics

- Fossil data sourced from the Natural History Museum Osteology Gallery.
- Quiz content developed with museum experts.
- No personal user data stored — only anonymized interaction metrics.

Implementation Details

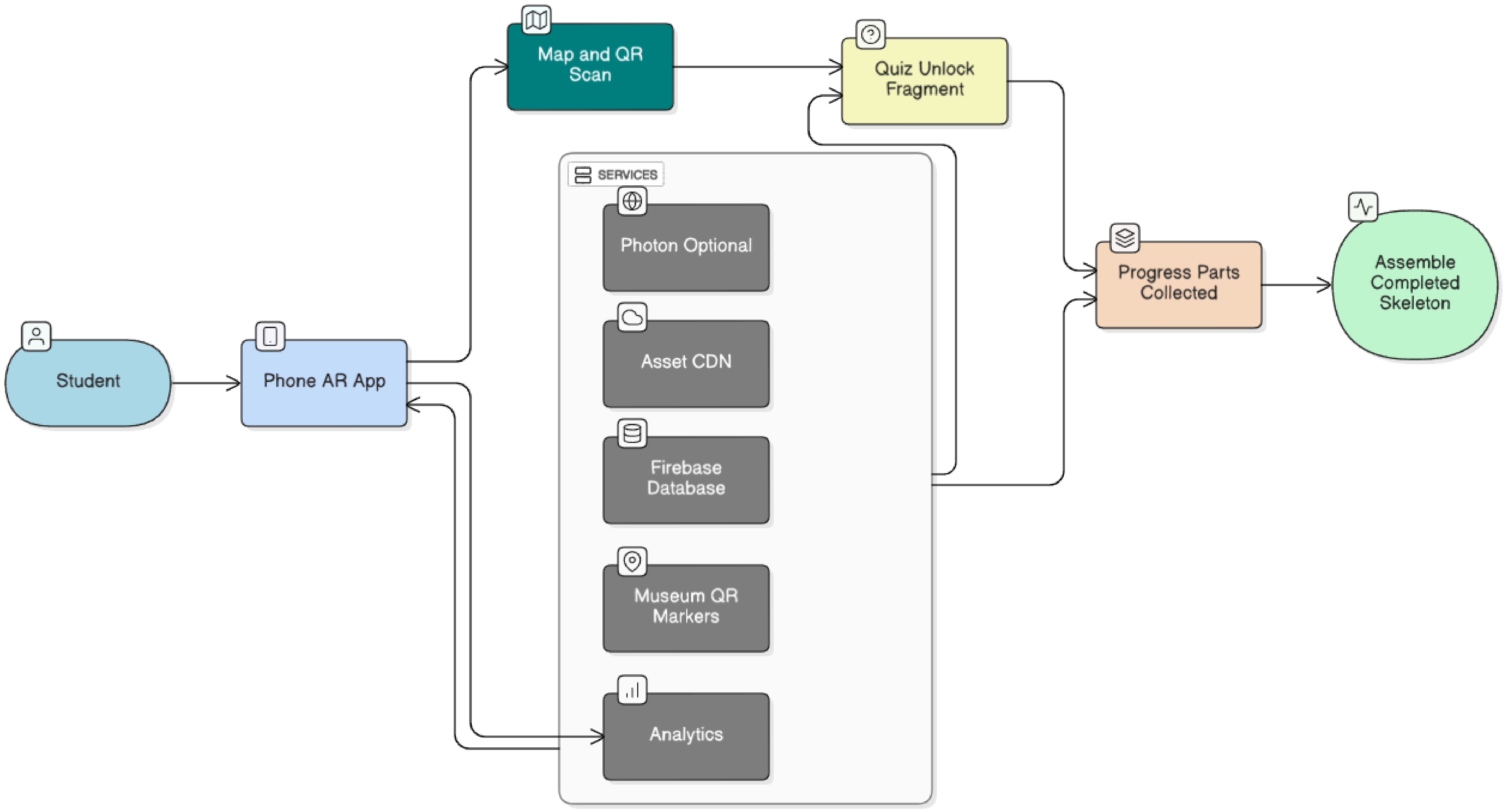
System Overview

- Visitors scan QR codes → Unlock fossil fragments → Answer quizzes → Collect pieces → Assemble full skeleton → View final AR animation.
- Treasure map guides exploration across exhibits safely and purposefully.

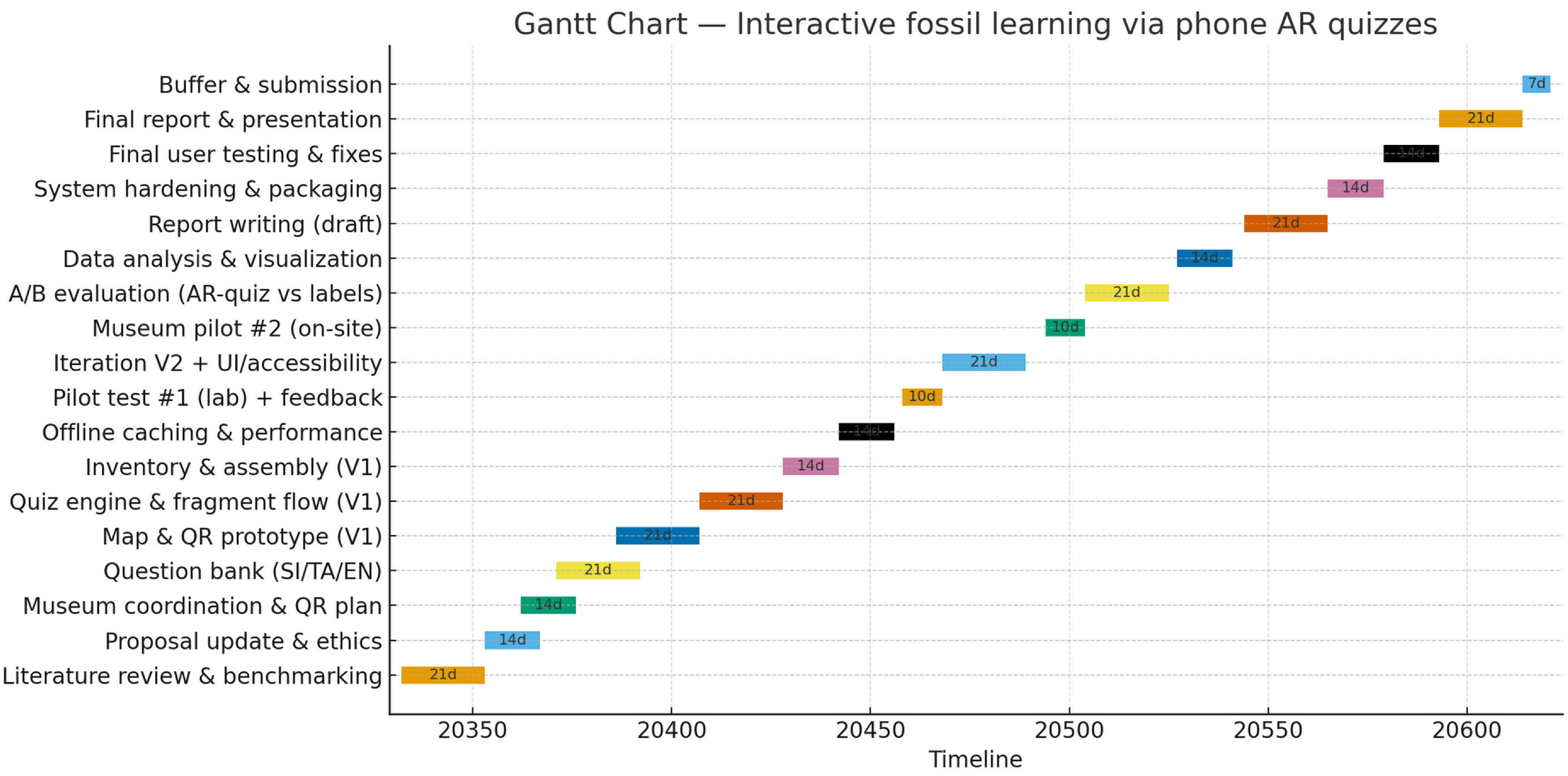
Real-World Use Case

- School groups visit the museum, collaborating to complete fossil skeletons.
- Families use the app for interactive learning, making museum visits fun and educational.
- Museum staff gain a tool to modernize exhibits and attract younger audiences.

Individual System Overview Diagram



Gantt chart



Gantt Chart — Interactive fossil learning via phone AR quizzes
Buffer & submission
7d
Final report & presentation
21d
Final user testing & fixes
System hardening & packaging
14d
Report writing (draft)
21d
Data analysis & visualization
14d
A/B evaluation (AR-quiz vs labels)
21d
Museum pilot #2 (on-site)
10d
Iteration V2 + UI/accessibility
21d
Pilot test #1 (lab) + feedback
10d
Offline caching & performance
Inventory & assembly (V1)
14d
Quiz engine & fragment flow (V1)
21d
Map & QR prototype (V1)
21d
Question bank (SI/TA/EN)
21d
Museum coordination & QR plan
14d
Proposal update & ethics
14d
Literature review & benchmarking
21d
20350
20400
20450
20500
20550
20600
Timeline